

Demonstrating Parallel Adaptation: How Switching between Two Virtual Bodies with Different Perspectives Enables Dual Motor Adaptation

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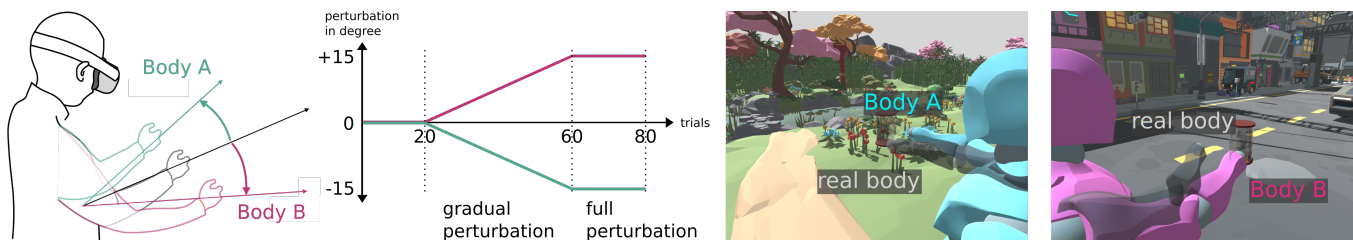


Figure 1: The elements of dual motor adaptation over 2 virtual bodies. From left to right: (1) the user motion perturbation depends on their VR body; (2) the perturbation increases over trials; (3) the body A in 3pp; and (4) the body B in 3pp. The dark arm in (3) and (4) represents the real arm position.

ABSTRACT

We demonstrate a dual-motor adaptation over 2 virtual bodies with opposite visuomotor rotation ($+15^\circ$ and -15°). A related study explored the effect of view perspective (1st person view perspective, 1pp, and 3rd person view perspective, 3pp) on a dual motor adaptation behavior. They showed that (1) in 1pp, the adaptation is explicit, while in 3pp, the adaptation is implicit, and (2) the implicit adaptation led to a better dual adaptation. Here we let the participants experiment with explicit and implicit dual motor adaptation by doing a reaching task in 1pp and then in 3pp. We present an ad-hoc motor adaptation score and discuss which adaptation behavior led to a better dual adaptation.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

motor adaptation, virtual bodies, multiple embodiment

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1 INTRODUCTION

In Virtual Reality (VR) we can embody a VR body with a non-standard body representation. Still, we know little about our ability to embody multiple VR bodies at the same time [Miura et al. 2021; Takada et al. 2021]. Can we even adapt our motion to 2 different bodies' internal models? Here, we demonstrate motor adaptation over 2 virtual bodies with opposite visuomotor rotation (VMR). By researching dual motor adaptations, we envision a VR system allowing us to embody multiple VR bodies.

In this demonstration, a participant experiments with 2 different types of motor adaptation (explicit and implicit) while switching between 2 VR bodies with an opposite VMR. After experimenting with them, the participant realizes that implicit adaptation allows for a better dual adaptation while maintaining a good level of body ownership and agency over the 2 bodies.

2 CURRENT RESULTS

This demonstration follows recent findings [Verhulst et al. 2022] regarding dual adaptations over 2 virtual bodies. We present some of those findings to help understand the demonstration reasoning.

We compared the direction errors (details after that) between a body with a 1st person view perspective (1pp) and a body with a 3rd person view perspective (3pp), in a setup very similar to the demonstration (see Sec. 3).

The direction error is the angle difference between the hand direction (the start position of the hand motion to the point of peak velocity of the hand motion) and the target direction (start-point position to end-point position).

The results (see Fig. 2) show that: (1) 1pp has a higher average direction error than 3pp in the full perturbation block (trials 50-90); (2) 1pp de-adaptation is "fast" and does not fit an exponential decay

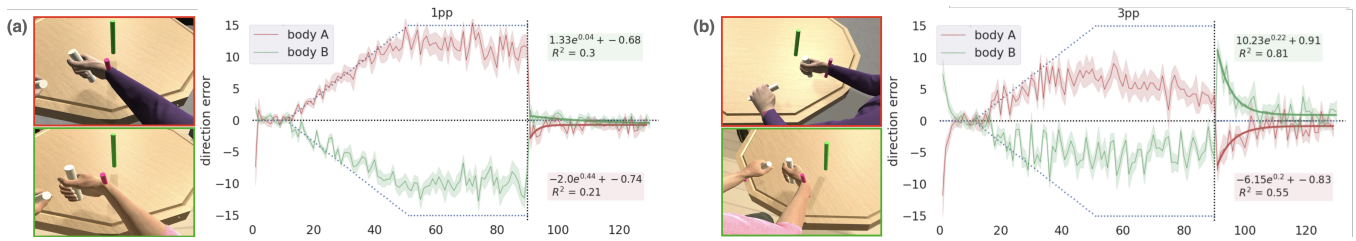


Figure 2: The direction error, with a correction offset, averaged by view perspective and bodies (body A in red, body B in green) across participants. (a) The perspective in the HMD and the direction error for 1pp and (b) for 3pp. Results from [Verhulst et al. 2022].

function, while 3pp is “slow” and fits an exponential decay function (resp. $R^2 \leq 0.3$ and $R^2 \geq 0.55$) in the washout block (trials 90-130). Therefore: 1pp shows key-elements of explicit adaptation (i.e., slow adaptation and fast de-adaptation), while 3pp shows key-elements of implicit adaptation (i.e., fast adaptation and slow de-adaptation) [Maresch et al. 2021].

We also compared differences in body ownership and agency before and during perturbation (using reported questionnaires). We found that 1pp led to a significant decrease in body ownership and agency when we added a perturbation, while 3pp led to a significant increase in body ownership. Also, for both 1pp and 3pp in the full perturbation block, the body ownership is rated average (with no sign. diff. between 1pp and 3pp), and the agency is high rated (with no sign. diff. between 1pp and 3pp).

3 THE DEMONSTRATION

The Material. We use a Meta Quest 2 HMD¹ and its 2 controllers.

VR Environment. We use 2 visually and spatially different scenes, 1 for each body (body A is in old-Japan, and body B is in neo-Japan, see Fig. 1). We also use another visually and spatially different scene (western-US) where the participant takes a break and de-adapts between 1pp and 3pp.

The Reaching Task. The participant starts from a static “start position” (represented by a sphere) and reaches a static “end position” (represented by a cylinder) with their dominant hand. When the participant reaches the target, they might (~50%) switch from their body to the other body (e.g., from body A in old-Japan to body B in neo-Japan). The order of the body is randomly pre-generated.

To facilitate the feed-forward reaching motion, the participant should reach the target in $\leq 700ms$. After 700 milliseconds, the cylinder’s color turns dark to signal the participant to move faster.

Both virtual bodies have a VMR along the participant’s dominant upper limb. The 2 perturbations start at 0° (trial 0-20), then increase over trials in the gradual perturbation block (trials 20-60), then keep their maximum values ($+15^\circ$ and -15° , trials 60-80) in the full perturbation block, see Fig 1.

Procedure. The participant starts the reaching task for 80 trials in 1pp, with a gradual VMR. Then the participant performs the task for 40 trials in 1pp without VMR to “washout” their motor adaptation, then waits 30 seconds. Then the participant starts the

task for 80 trials in 3pp. After completion of the 2 reaching tasks, the participant sees their score for 1pp and 3pp (details after that). The ad-hoc score is calculated as 360 minus the sum of the absolute values of the average direction error e for body A and body B, in the full perturbation block (trials 60-80): $360 - |e_A| + |e_B|$. We use a score because (1) it gives an easy idea of the best overall adaptation (higher better means better adaptation) and (2) to gamify the task.

4 CONCLUSION

During the procedure in 3pp, we record the participant’s dominant hand motion from a top-down view, with a camera fixed above the participant. After the participant removes the HMD, we play the recording so they can see the differences in motion between body A and body B (since, in implicit adaptation, they likely do not have the feeling of making different motions and need to see them after the fact).

According to our results (see Sec. 2 and Fig. 2), the participants will have a better score in 3pp (thanks to the implicit adaptation). The demonstration will let the participants experiment with a dual adaptation over 2 virtual bodies with opposite VMR and might bring them closer to a feeling of parallel embodiment.

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¹<https://store.facebook.com/jp/quest/products/quest-2>